

Control-M SaaS Operations

Day-to-day monitoring and incident handling: viewpoints, job statuses, job actions, alerts, services, and troubleshooting runbooks.

1. The operator's day in SaaS

With BMC running the control layer, operations focuses on the workload itself: jobs, Agents and the applications they touch. You no longer restart Control-M/Server services or watch the EM database. If the SaaS platform itself has a problem, you raise it with BMC Support and follow the BMC status page.

When	What to check
Start of shift	Handover notes; Alerts window; jobs in Ended Not OK, Wait User and Hold; Agent availability
After New Day	Expected folders ordered; nothing stuck in Wait Scheduling; holiday calendars applied correctly
Through the shift	Critical services on track; long-running jobs vs. their averages; late-submission alerts
End of shift	Open incidents and any manual actions taken (with reasons) in the handover

2. Monitoring domain basics

- **Viewpoints:** filtered views of active jobs, e.g. by application, folder, status or host. Build one per team and one 'all failures' view.
- **Services:** business-level view of a flow with an expected completion time; shows On Time, Late or Failed.
- **Alerts:** messages raised by jobs (Notify to Alerts window) and by the system. Handle them, note what you did, then mark them handled.
- **Job Search:** find jobs ordered today or on earlier dates by name, run ID or a file name the job references.
- **Neighborhood:** show predecessors and successors of a job, with a radius. Use it before any action that affects a flow.

Job statuses

Status	Meaning	Usual next step
Wait Scheduling / Wait Time	Not yet in its time window	None, unless it should already have run
Wait Event	Waiting for a predecessor event	Check Waiting Info and the predecessor
Wait Resource	Lock or pool resource not available	Find which job holds it
Wait Host	Agent / host not available	Check the Agent (ping), then the server
Wait User	Needs operator confirmation	Confirm if approved
Executing	Running on the Agent	Compare elapsed time to the average
Ended OK	Completed successfully	None
Ended Not OK	Failed, or set Not OK	Read Output and Log, then fix or escalate
Held	Paused by an operator	Release when the reason is cleared
Unknown / Disappeared	Status lost, often after an Agent outage	See runbook 4.3

Job details tabs

Tab	Use it to
Summary	Start/end times, run ID, run date, host, run-as user, estimated start and end
Waiting Info	See exactly which events, resources or predecessors a job is waiting for
Log	See every status change and action on this run, including who did what
Output	Read the job's sysout to find the actual error
Statistics	Average run time and CPU, used for 'is this job slow?' questions
Script	View the script or command; editing needs the Edit JCL/Script authorization
Related / SLA	Which services depend on this job

3. Job actions: what each one really does

Action	Effect	Use when
Hold / Release	Stops the job from starting (or pauses it) / lets it continue	You need to edit it or delay it safely
Rerun	Runs a job that has already finished again	Root cause fixed and the job is safe to repeat
Run (Run Now)	Starts it immediately, ignoring its submission criteria	Business asks for an urgent run
Confirm	Releases a job waiting for user confirmation	Approval received
Kill	Terminates a running job (Agent must be available)	Job is hung or must be stopped
Set to OK	Marks Ended OK and runs post-processing, so successors start	Work was done another way, with approval
Not OK (No PP)	Marks Ended Not OK, frees its resources, skips post-processing	Job stuck in Executing / Unknown and the Agent is down
Skip	Doesn't run the job; lets successors continue	Approved to bypass this step today
Bypass Options	Ignore time limits, events, resources, or run as Dummy / without post-processing	Targeted override of one blocker
Apply Events	Adds the missing events from Waiting Info	Predecessor was handled manually
Edit (Monitoring)	Changes this run only, not the Planning definition	One-off fix to today's run
Delete / Restore	Removes a held job from today's run / brings it back	Job ordered by mistake
Run Jobs/Folders	Order jobs or folders on demand with run attributes	Ad-hoc or re-run of a whole flow

Before forcing a job

Set to OK, Skip and Bypass make successors run on the assumption that the work is done. Get approval from the application owner, check the Neighborhood first, and enter a clear annotation.

Run Jobs/Folders attributes

- **Ignore Scheduling Criteria:** order it now even if today isn't a scheduled day.
- **Run as Independent Flow:** adds a unique suffix to events so this instance doesn't interfere with today's regular run.
- **Hold:** order it in Held state so you can check or edit before it starts.
- **Date Selection:** current working date, a different run date (sets %%ODATE), or wait for a date.
- **Place in Folder:** New instance, Standalone, Recent, or a Selected run ID.
- **Variables:** override or add variables for this run only.

4. Runbooks

4.1 Job ended Not OK

1. Open **Output** and find the first real error, not just the last line.
2. Check **Log**: is this the first failure or a repeat? Did an automatic rerun already happen?

3. Check **Statistics** and recent history: did it fail the same way yesterday?
4. Classify the failure: data or business issue (application team), infrastructure (host, disk, network, DB), credentials or connection profile, or a definition problem (scheduling team).
5. Fix or escalate per the job's documentation link. After the fix, **Rerun**. If the step must be bypassed, get approval, then Set to OK or Skip with an annotation.
6. Record the incident number and action in the alert and the handover.

4.2 Job waiting too long

1. Open **Waiting Info** to see the exact blocker.
2. **Wait Event**: find the predecessor (Neighborhood) and fix that job first. Check for an ODAT/PREV date mismatch on the event.
3. **Wait Resource**: find which job holds the lock or pool units. It may be hung.
4. **Wait Host**: ping the Agent; see 4.3.
5. **Wait User**: confirm only with approval.
6. Use Apply Events or Bypass Options only once the real dependency is known to be satisfied.

4.3 Agent unavailable

1. Configuration > Agents: confirm the status and **Ping** the Agent.
2. On the host: is the server up? Is the Agent service running? (Windows: the Control-M/Agent service in services.msc.)
3. Restart the Agent service if it is stopped, then ping again from the web interface.
4. Check outbound connectivity from the host to the SaaS endpoints (proxy, firewall, DNS, certificate inspection).
5. If jobs were Executing when the Agent went down: once the Agent returns they usually reconcile. If the Agent can't come back and the work must move on, use **Not OK (No PP)** to release resources, then rerun on another host in the host group.
6. If many Agents drop at once, suspect a network or SaaS-side issue and open a BMC case.

```
# Windows (elevated PowerShell) on the Agent host
Get-Service | Where-Object DisplayName -like '*Control-M*'
Restart-Service -DisplayName 'Control-M/Agent*'
Test-NetConnection <your-tenant-endpoint> -Port 443

# Agent-side diagnostics (from the Agent's exe/scripts directory)
ag_ping
ag_diag_comm
```

4.4 Job running much longer than usual

1. Compare elapsed time with the average in Statistics.
2. Look at the host: CPU, memory, disk, blocking in the database, or the process waiting on a file or lock.
3. Ask the application team before killing. A partial kill can leave data half-processed.
4. If approved, **Kill**, fix the cause, then **Rerun**. Note any successor impact on services.

4.5 New Day issues

- Expected folder not ordered: check its calendar for today, Activity Period, and whether the latest definition was checked in.

- Ordered on a holiday: check the confirmation calendar and Exception Policy, and whether next year's holiday dates were loaded.
- Order manually with Run Jobs/Folders and the correct run date if business needs it.

5. Automation API for operators

```
ctm run jobs:status::get -s "application=FIN&status=Ended Not OK"
ctm run job:output::get <jobId>
ctm run job:log::get <jobId>
ctm run job:rerun <jobId>
ctm run job:hold <jobId> | ctm run job:free <jobId>
ctm run job:confirm <jobId> | ctm run job:setToOk <jobId>
ctm run order <server> <folder> [<job>]
```

These make good building blocks for scripted health checks, for example a PowerShell report every morning of all failed or held jobs per application, sent to the team channel.

6. Escalation matrix

Situation	Owner
Job logic, data, or application error	Application support team
Host down, disk full, service account locked	Infrastructure / Windows or Unix team
Definition, calendar or dependency wrong	Scheduling team (Planning)
Agent install, plug-in, connection profile, access	Control-M administrators
Web UI unavailable, many Agents disconnected, SaaS errors	BMC Support case (plus internal major-incident process)

Golden rules

- Read the output before acting.
- Check the Neighborhood before forcing anything.
- Never Set to OK or Skip without approval, and always annotate.
- Edit in Monitoring fixes today only. Raise a change for the Planning definition too.
- Write it in the handover.

7. Setting up your monitoring

Viewpoints that work

Viewpoint	Filter	Who uses it
All Problems	Status = Ended Not OK, Wait User, Held, Unknown	Every operator, always open
Critical Path	Jobs marked Critical, or jobs in key services	Shift lead
By Application	Application = FIN / PAY / HR ...	Operators assigned to that application
Long Runners	Status = Executing (compare with averages)	Shift lead, during the night
Waiting Too Long	Wait Event / Wait Resource / Wait Host	Operators, after main batch start
Today's Manual Actions	Jobs with annotations or manual status changes	Handover and audit

- Keep viewpoints small and focused. A viewpoint with 20,000 jobs hides the five that matter.
- Save shared viewpoints for the team so everyone sees the same thing.
- Use the flow (graph) view to understand dependencies and the list view to sort and act quickly.

Alert handling procedure

1. Read the alert, the job name, and the urgency (Normal, Urgent, Critical).
2. Acknowledge or assign it so colleagues don't work the same alert.
3. Open the job and follow the matching runbook in section 4 or 8.
4. Record the action and any incident number in the alert note.
5. Mark the alert handled only when the job is recovered or formally handed to the owning team.

Priority guide

Priority	Definition	Target response
P1	Critical service will miss its deadline, or many jobs/Agents down	Immediate; call the on-call owner; major-incident process
P2	Critical job failed but there is time before the deadline	Within 15 minutes
P3	Non-critical job failed; no business deadline today	Within the shift; ticket to the owning team
P4	Warning only (e.g. slow run that still finished)	Note in handover; trend review

8. More runbooks

8.1 File transfer (MFT) job failed

1. Open the job and use **Show File Transfers**, or search in the Managed File Transfer domain by file name or run ID.
2. Read the transfer status: Failed, Completed with Errors, Canceled.
3. Common causes: remote host down, authentication failure (expired password or key), host key changed, file not present, permission denied at the destination, disk full.
4. Host key changed: do not just accept the new key. Confirm with the partner that they changed it.
5. Authentication: the administrator updates the connection profile; then rerun.

6. File not present: check the agreed time with the partner; use Keep Active or rerun later if the file is late.

8.2 Database job failed

- Read the database error in the output (login failed, timeout, deadlock, constraint violation).
- Login failed: password or account lock on the connection profile. Escalate to the admin and DBA.
- Deadlock or timeout: check for a conflicting job; a lock resource may be needed to serialize them.
- Data errors (constraint, conversion): the application team owns the fix. Do not rerun blindly.

8.3 Job ran twice / duplicate processing

- Check the Log: was it a manual rerun, an automatic If-action rerun, or a second order of the folder?
- Check whether the successor waits for an event that was not deleted after the first run.
- Check Run Jobs/Folders history: someone may have ordered the folder again without 'Run as Independent Flow'.
- Inform the application team immediately. Duplicate loads may need a data clean-up.

8.4 Lock or resource wait that never ends

1. Open Waiting Info on the waiting job to see the resource name.
2. Find which job holds the lock: search for jobs using that resource, usually in Executing or a hung state.
3. If the holder is genuinely running, wait or escalate according to its expected end time.
4. If the holder ended but still shows as holding (for example after an Agent failure), use **Not OK (No PP)** on it to release the resource, with approval.
5. As a last resort, use Bypass Options > Lock Resources on the waiting job, only after confirming the two jobs cannot conflict.

8.5 Cyclic job stuck or not cycling

- Check Summary > Estimated Run Times and Rerun Every to see when the next cycle is due.
- If it ended Not OK, the cycle normally stops; rerun it after fixing the cause.
- If it reached Maximum Reruns, it has finished for the day. Re-order only if business needs more runs.
- If it is past its To time, it will not cycle again today.

8.6 File watcher still waiting

- Check the exact path and pattern in the job against the actual file name (case, date format, extension).
- Check that the run-as account can see the folder (UNC path, share permissions).
- Contact the sending team: has the file been sent? Was it written under a temporary name?
- If the file arrived under a different name, ask the scheduling team before renaming files by hand.

8.7 Many jobs failing at once

1. Look for the common factor: same host or host group, same application, same connection profile, same time.
2. Same host: Agent or server problem; see runbook 4.3.
3. Same database or SFTP profile: endpoint down or credentials changed.

4. Across everything: suspect a network, DNS or SaaS problem. Check the BMC status page and open a P1.
5. Consider holding the affected folders to stop more failures while the cause is fixed, then release them in order.

9. Windows exit codes and messages

Control-M decides Ended OK or Not OK from the exit code (0 = OK unless rules say otherwise). Some useful Windows codes:

Exit code / message	Meaning	Typical action
0	Success	None
1	Generic failure; for robocopy it means files copied	Read output; robocopy 1-7 is normally success
2	File not found (many tools)	Check path and file name
3	Path not found	Check the folder exists and is reachable (UNC)
5	Access denied	Run-as account permissions on the file, folder or share
1326	Logon failure: bad user name or password	Run-as or connection credentials changed
1909	Account locked out	Unlock the service account; find what is using the old password
-1 / 4294967295	Script terminated abnormally (often unhandled PowerShell error)	Read output; fix the script's error handling
'is not recognized as an internal or external command'	Program not in PATH for the service account	Use the full path to the executable

Tip for script owners

PowerShell scripts should end with an explicit `exit 0` or `exit 1`. Without it, a script can report success even when a command inside it failed, and Control-M will show Ended OK.

10. Scripted health checks

The Automation API REST interface uses the `x-api-key` header. Use a token whose role can only read job status for reporting scripts.

```
# morning_report.ps1 - list failed and held jobs per application
$endpoint = 'https://<your-endpoint>/automation-api'
$headers = @{ 'x-api-key' = $env:CTM_API_TOKEN }

$failed = Invoke-RestMethod -Headers $headers `
    -Uri "$endpoint/run/jobs/status?status=Ended%20Not%20OK&limit=1000"
$held = Invoke-RestMethod -Headers $headers `
    -Uri "$endpoint/run/jobs/status?held=true&limit=1000"

$failed.statuses | Group-Object application |
    Select-Object Name, Count | Sort-Object Count -Descending | Format-Table

$held.statuses | Select-Object name, folder, application, host |
    Format-Table -AutoSize
```



```
# Check that every Agent on a server is available
$agents = Invoke-RestMethod -Headers $headers `
    -Uri "$endpoint/config/server/<server>/agents"
$agents.agents | Where-Object status -ne 'Available' |
    Select-Object nodeid, status
```

Schedule these scripts as Control-M jobs themselves. Then the report arrives every morning and a failed report is itself visible in Monitoring. Confirm the exact query parameters and response fields against the Run service reference for your API version.

11. AI features in Control-M SaaS

Recent SaaS releases include AI capabilities under **Control-M AI**. What is enabled depends on your tenant and licensing.

Feature	What it helps with in operations
Jett AI assistant	Ask questions in plain language about jobs, failures and the environment
Execution Insights	Explanations of why a job failed or behaved unusually, based on its output and history
AI Pilot for Monitoring	Assistance with monitoring tasks and suggested next steps
AI Pilot for Planning / Workflow Creator	Help for schedulers building workflows (less relevant for L1)

Use with care

Treat AI suggestions like advice from a colleague: verify against the job output and the runbook before taking actions such as Set to OK or Kill.

12. Templates

Shift handover

SHIFT HANDOVER Date: _____ Shift: ____ From: _____ To: _____

- Overall status: Green / Amber / Red
- Critical services: FIN reports ____ PAY batch ____ HR ____
- Open failures:

Job	Folder	Status	Incident #	Owner	Next action	ETA
- Manual actions taken (Set to OK / Skip / Kill / Bypass / Edit):

Job	Action	Approved by	Reason	Time
- Agents / hosts unavailable:
- Jobs on Hold and why:
- Expected late files / known issues for next shift:
- BMC cases open:

Incident ticket

Summary : [CTM] <JOBNAME> ended Not OK - <short error>
Job : <folder>/<jobname> Run ID: <orderid> ODATE: <date>
Host : <agent / host group> Run as: <account>
Time : started <hh:mm>, failed <hh:mm> IST
Error : <first real error line from Output>
Impact : <service / report / downstream jobs affected, deadline>
Actions : <what L1/L2 already did: rerun, checks>
Request : <what the owning team needs to do>

Approval request for a forced action

Hi <owner>, job <JOBNAME> (ODATE <date>) is <status> because <reason>.
To let <successor / service> continue, we propose <Set to OK | Skip | Bypass>.
Impact: <what will run next and on what assumption>.
Please confirm by reply; we will record your approval in the job annotation.

13. Daily operations checklist

Time (IST)	Check
Shift start	Read handover; open All Problems and Critical Path viewpoints; check Alerts
After New Day	Expected folders ordered; holiday logic correct; nothing in Wait Scheduling that should run
Batch start (e.g. 00:00-02:00)	Agents available on all key host groups; file watchers active
Every hour	All Problems viewpoint empty or every item owned; services On Time
Before deadlines	Critical services; long runners vs. averages; escalate early if At Risk
Shift end	Handover complete; manual actions annotated; tickets updated

14. Frequently asked questions

Question	Answer
Rerun vs. Run (Run Now)?	Rerun repeats a finished job. Run starts a waiting job immediately, ignoring its submission criteria
Set to OK vs. Skip?	Set to OK marks a job done and runs its post-processing. Skip doesn't run it and lets successors continue
Why can't I kill the job?	Kill needs the Agent. If the Agent is down, use Not OK (No PP) with approval
I edited a job in Monitoring; will it be the same tomorrow?	No. Monitoring edits affect this run only; the Planning definition must be changed too
The job ran but did nothing and ended OK?	The script probably doesn't return a failure exit code. Raise it with the script owner
Why is the job's date yesterday?	It belongs to the previous logical day (ODATE) and is still active, e.g. through Keep Active
Who restarts the Control-M server?	Nobody on your side. In SaaS, BMC runs the server; open a BMC case for platform issues

15. Glossary

Term	Meaning
Annotation	Reason entered by an operator when taking an action; kept for audit
Neighborhood	A job's predecessors and successors
Post-processing	If-actions, events, notifications and output handling after a job ends
Run ID / Order ID	Identifier of one execution instance of a job or folder
Run count	How many times this instance has run (increases with reruns)
Service	Business-level group of jobs with a deadline, shown in Monitoring
Viewpoint	Saved, filtered view of active jobs
Waiting Info	Tab showing exactly what a job is waiting for

Based on BMC's Control-M SaaS documentation (documents.bmc.com/supportu/controlm-saas), checked October 2026. Control-M SaaS was previously called BMC Helix Control-M. Menu names and limits change between releases, so confirm details against your tenant and the current docs.